

measure the relative positions of the minutiae; much the same way that people used to navigate in ancient times by using relative positions of stars in the sky to get their bearings.

Depending on the algorithm, a specific number of minutiae must be matched for the print to be accepted. Despite the various sums and checks in the system, fingerprint scanning is not even remotely a foolproof system. If you look around on the Internet, you will find a truckload of sites giving detailed instructions on how to go about faking fingerprints! In fact, in a recent study at Yokohama National University, Japan, scientists easily fooled even the latest fingerprint systems using fake prints made out of gelatine.

Retinal Scanning

Despite being shown as the absolute cutting edge, retinal scanning is actually rather old in the chronology of technology innovations and research on this started way back in the 1930s. For the retina to be scanned, the user looks through a small hole in the scanning device and focuses on a particular point for the time period during which, a low intensity light and a CCD analyse the layer of blood vessels at the back of the eye for matching patterns (akin to fingerprint checking) and validate or repudiate the persons identity.

This technology is still not in the public domain (unlike fingerprint recognition, which is) and is used only to secure highly sensitive security areas.

Unlike fingerprints, there is absolutely no known method of replicating a person's retina and to use a dead person's retina is no good as it deteriorates too fast to be of any help.

Iris Scan

Iris Scan, though relating to the eye (like retinal scan) uses a completely different method of identification. The Iris is the coloured ring surrounding the pupil. The scan analyses the features that exist in this coloured tissue.

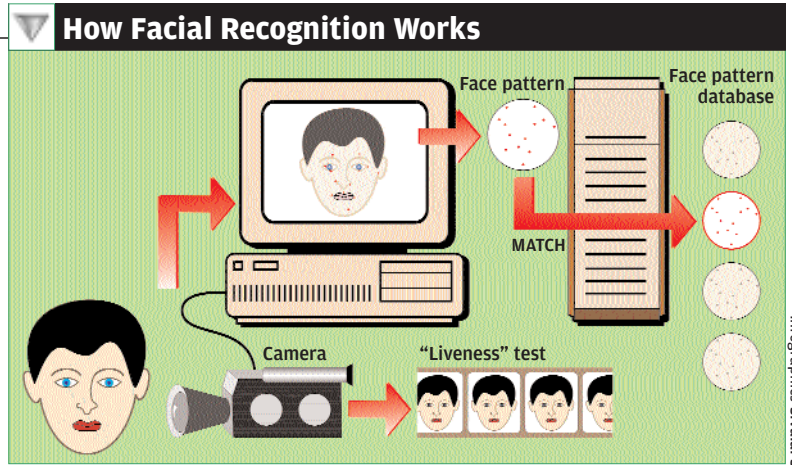
Over 200 points can be used for comparison such as the rings, furrows and freckles. The scan is done with a regular camera and the subject stands about a foot from the lens (of the camera) so it is a lot more convenient.

The Iris pattern is much more unique than a fingerprint. A statistical analysis puts the probability of two irises matching at 1 in 10 to the power 78 while the population of humans on earth is roughly 7 billion that is 7 to the power 9.

Facial Recognition

While fingerprinting and retinal scanning are relatively easy to administer, since the people going through the process are aware of it and are consenting to subject themselves to these measures, the main application of facial recognition is in security wherein the software is expected to pick a face out of, say, thousands of passengers at the airport, and match it with a database of wanted criminals and positively state whether or not that face belongs to the guilty party.

To make the computer recognise a face from a picture or a video feed is quite an achievement in itself, but a bigger achievement is to



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identify clearly if the face is that of the wanted man or not.

If you look in the mirror, or at a person's face for that matter, you will notice that every face has certain characteristics and distinguishable features, which allow us to differentiate between two people. The equipment used here is not really too fancy or cutting edge, and the brains for this lie entirely in the software.

The software divides the face into 80 nodes, some of the common ones being distance between eyes, width of nose, and depth of eye sockets, cheekbones, jaw line, and chin. The system generally needs to match between 14-25 nodes in order to obtain a positive ID.

Now, obviously, there are a lot of people coming in and out of a place where this system is set up (stadiums, airports etc). The real challenge is to recognise a face instantaneously. To facilitate this, a database is created with the help of an algorithm, which goes through the characteristics of the faces and stores them as a string of numbers. This string is called a face print.

The following are the broad steps utilised by facial recognition software.

☐ **Face Detection:** The camera pans around looking for a face. The minute it encounters a face, it starts scanning it and proceeds to identifying the various nodes and taking measurements if possible

☐ **Detection of Orientation:** Once the face is detected, the system determines the head's size and position. Generally, a face needs to be around 40 degrees towards the camera for the system to register and analyse it

☐ **Mapping:** The facial image is scaled down to the level of the images in the database and is then rotated and otherwise adjusted to match the formatting of the images in the database.

☐ **Encoding:** The algorithm then converts the face into a face print based on the pre-defined criteria programmed into the algorithm.

☐ **Matching:** This new data is then used as a filter to sort through the database of faces at super fast speeds to come up with a match.

Since it uses a variety of nodes, simple alterations of the face will not fool it; however, twins might; so the system is certainly not infallible.

Voice Recognition

Like fingerprints and face attributes, every person has a unique speech pattern. Voice recognition works by first storing voice pat-